

Contemporary C++: *Learning Modern C++ in a Modern Way*



مدرس: سعید امراللہی بیوکی

Saeed Amrollahi Boyouki

www.Linkedin.com/in/saeed-amrollahi-boyouki

Agenda 20/30

Session 20. More on Classes, Access Controls, and Introduction to Constructors

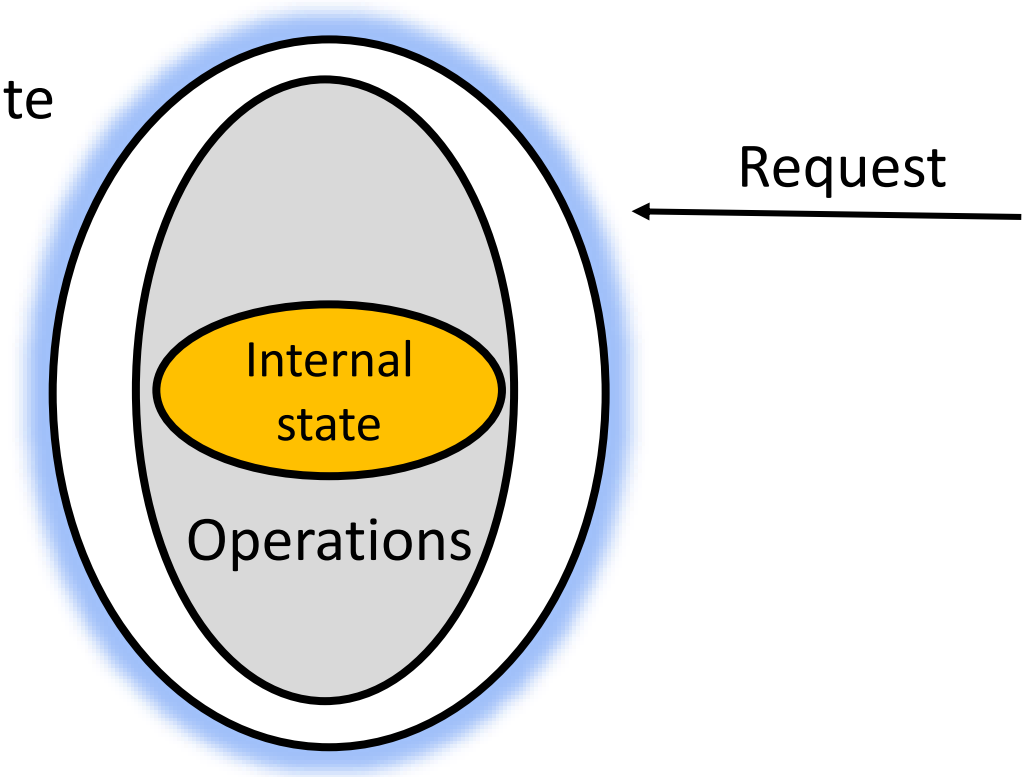
- 📊 Fundamental types vs. User-defined types
- 📊 The C++ classes classification: Concrete and Handle classes
- 📊 Classes: Data members and Member functions
- 📊 References and further readings
- 📊 Q&A

120 min (incl. Q & A)



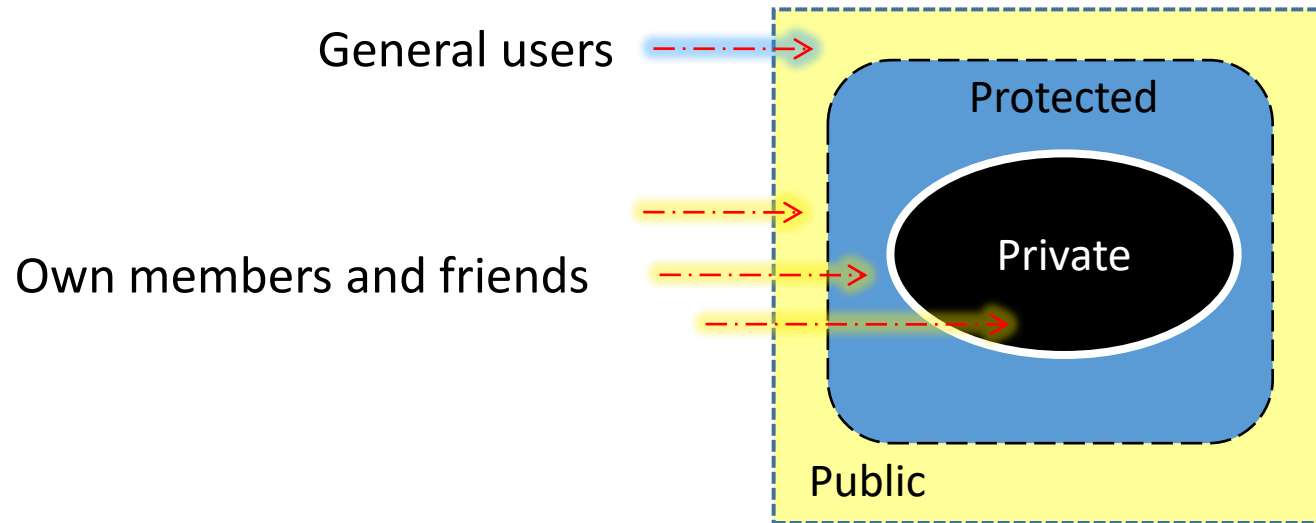
Class/object **D**esign: fundamental model

- The egg model
 - Requests are the *only* way to get and object to execute an operation.
 - Operations are the *only* way to change an object's internal data (state).
- Example: The car, the stack



Access control

- A member of a class can be *private*, *protected*, or *public*.
- If it is *private*, its name can be used only by members and friends of the class in which it is declared.
- If it is *public*, its name can be used anywhere without access restriction.



- Access to members vs. Member visibility

Access Control, class scope and resolution operator

- (Non-inline) member functions definitions comes in source file.

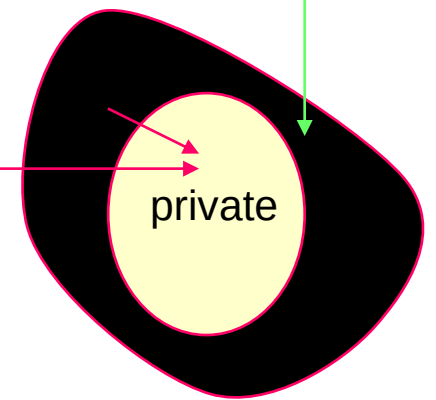
```
// date.cpp
void Date::add_year(int n)
{
    year += n;
}
void Date::add_month(int m)
{
    // ...
}

void Date::add_day(int d)
{
    // ...
}
```

Scope resolution operator

own member
functions and
friends

General users



- Nonmember functions are barred from using private members.

```
void f()
{
    Date d;
    d.init(18, 8, 2023);
    d.Day++; // error: day is private
    Date dd;
    dd.init(36, 13, 1380); // invalid date
}

void Hijack(Date& d)
{
    d.Month--; // error: can't access to private member
}
```

- Local scope
- Function Scope
- **Class Scope:** A class is a namespace containing its members.
- Namespace Scope

Constructor

- Constructors are like “init” functions.
- Constructor aka ctor

A ctor is used to *initialize* objects of its class type.

A ctor establishes a *local environment* in which member functions execute.

Constructor

A ctor is often used to establish an *invariant* for the class.

A ctor is often used to acquire *resources*.

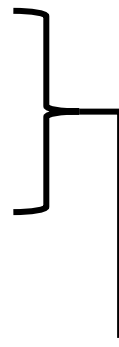
- A constructor is recognized by having the same name as the class itself. In other words, constructors do not have names. Constructors do not return value.

Constructor: Initializing objects

```
class Date {  
    // ...  
public:  
    // ctor declaration  
    Date(int, int, int); // constructor  
};  
Date d1 = Date(1, 1, 2019);  
Date d2(1, 1, 2019);  
Date d3; // error
```

- Multiple constructors: Several ways to initialize an object
- Overloaded constructors: Function name overloading

```
class Date {  
    // ...  
    Date(int, int, int); // day, month, year  
    Date(int, int); // day, month, today's year  
    Date(int); // day, today's month and year  
    Date(); // default Date: today (global)  
    Date(const char*); // Date in string rep.  
};
```



```
class Date {  
    // ...  
    Date(int = 0, int = 0, int = 0); // day, month, year  
    Date(const char*); // Date in string rep.  
};
```

- Default arguments

The **V**ector Class

- The very 1st try: fixed-size and struct-based vector

The Date Class cont.

- Function declared within a class definition are called *member functions*.
- Scope resolution operator: ::

```
void Date::init(int d, int m, int y)
{
    day = d; month = m; year = y;
}
// use date
Date birthday; // uninitialized object
Date today; // another object
Date unix_time_epoch; // Unix epoch time
today.init(1, 11, 2018); // initialize today
unix_time_epoch.init(1, 1, 1970); // initialize epoch time
```

3. Access Control and Information Hiding

```
// date.h
class Date {
    // data members are encapsulated
    int Day, Month, Year;
public:
    // member functions
    void init(int, int, int);
    void add_year(int);
    void add_month(int n);
    void add_day(int n);
    // ...
};
```

Implementation

Interface

own member functions and friends

General users

public

private

class declaration/definition

```
class X { ... };
```

- The class definition comes in a header file → component fundamental rule

R eferences and further readings

[Str13] Bjarne Stroustrup. The C++ Programming Language. 4th edition. Addison Wesley. Reading Mass. USA. May 2013.

Thanks for your patience ...

A man who asks a question is a fool for minute,
The man who does not ask, is a fool for a life.
- Confucius

Learning to ask the right (often hard) questions is an essential part of learning to think as a programmer.

- Bjarne Stroustrup *programming Principles and Practice Using C++, page 4.*

There is no stupid question, but there is stupid answer.
- Howard Hinnant

